NAME - VANSH TANWAR

ROLL NO - 2401010102

DBMS ASSIGNMENT 4

THEORY

SUBMITTED TO : Mansi Kajal Maam.

DBMS
Assignment = 4

Section- A

Q1- What is transaction in DBMS?

Ans A transaction is a logical unit of work that consist of one or more database operations. (eg. read, write, update). It must be executed in its entirely or not at all, preserving the integrity of the database.

Q2. Explain ACID properties.

Ans. # Automicity- Transaction is treated as a single, indivisible unit, either all operation are committed.

Consistency- Transaction brings the database from one valid state to another, preserving all defined rules.

Isolation- Concurrent execution of transaction results in state that is equivalent to some serial execution.

Durability- Once a transaction is committed, its changes persist even after a system failure.

Q3. What is Concurrency Control?

Ans Concurrency control is the process of managing simultaneous transaction so that they do not interfere with each other, preventing anomalies like lost update, dirty reads. and inconsistent analysis.

Q4 What is a deadlock?

Ans A deadlock is a situation where two or more transactions are waiting indefinitely for each other to release locks. for eg., Transaction T1 holds lock A and wait for B, while T2 holds lock on B and waits for A.

Q5. What is serializability?

Ans Serializability is a property of a concurrent schedule that guarantees its outcomes is the same as some serial execution of the same transaction.

Q6. What is recovery in DBMS?

Ans. Recovery in the DBMS is the mechanism that stores the database to a consistent state after a failure eg. system crash.

Q7. What is checkpoints?

Ans A checkpoint is a point in time where the DBMS saves the current state of the database. It reduces recovery time by limiting the log that must be processed after a failure.

Section B.

Design, Analyze & Defend.

Q1 Explain ACID properties using banking examples.

Ans. • Atomicity: Transfer of ₹1000 from account A to B deduct from A to add to B. If the second operation fails, the whole transaction is rolled back; no partial update

• Consistency: Before and after the transfer; total balance of A+B remains the same, and the both balances remain non-negative

• Isolation: Two concurrent transfers between the same account appear as if they run one after another; no intermediate dirty balance is seen.

• Durability: Once the transfer is commited the new balance survives a power outage.

Illustrate a deadlock situation with 2 transaction

Transaction T1: Lock X ; Lock Y;
Transaction T2: Lock Y; Lock X;
If T1 locks X and T2 locks Y simultaneously then T1 waits for Y (held by T2) and T2 waits for X (held by T1) - deadlock.

Explain serial vs concurrent execution with examples.

* Serial execution - T1 runs fully, then T2 runs
Example T1: read A, write A; T2: read A write A. Result is deterministic

• Concurrent execution - Interleaved operations
Example T1 reads A, T2 reads A, T1 writes A, T2 writes A - may cause lost updates.

Proper concurrency control (e.g., locking) making it equivalent to serial.

Sec - C (Case Based)

Ques A banking system transfer money between accounts. Explain how ACID properties apply.

Ans Atomicity - Deduction from source and addition to destination happen together or not at all.

- Consistency - Total balance of both accounts unchanged, no negative balances.
- Isolation - A third transaction seeing the transfer see either the full old state or full new state, never an intermediate state.
- Durability - After 'commit' the transfer persist even if the system crashes immediately.

Ques Two transaction update same data simultaneously. Identify possible issue.

Ans Lost update - One transaction's write overwrite the other's without proper locking.

- Dirty read - One transaction read uncommitted data written by other which may later rollback.
- Unrepeatable read - A value change between two read within the same transaction.
- Phantom read - New rows appear or disappear between two queries.

Ques A system crashes during transaction. Explain recovery steps.

Ans After crash, DBMS restarts and examines the log.

- Undo all transaction that are not yet committed.
- Redo all transaction that have committed but whose changes may not have been written to disk.
- Use the last checkpoint to limit log scanning.
- Database is restored to a consistent state.

Sec - D.

Explain one ACID property with real-life example
* Isolation :
→ Two people booking the last seat on a flight using the same website. Isolation ensures that only one booking succeeds; the other sees the seat already taken as if they booked one after another, not simultaneously.

Draw a transaction timeline and identify conflict :
→ Timeline :
T1 : Read (A) → Write (A) → Commit
T2 : Read (A) → Write (A) → Commit
Interleaved : T1 Read (A), T2 Read (A), T1 Write (A), T2 Write (A) → Conflict : T2's Write (A) conflicts wit T1's Write (A) (Write-Write conflict). Also T2's Read(A) after T1's Write (A) would be read write conflict.

Explain one recovery technique.
- Log-based recovery (Write-Ahead Logging) → Every transaction action is recorded in a log (undo/redo records) before the actual database update. On failure, the log is used to undo incomplete transactions and redo committed ones. Checkpoints reduce log volume.